

Cv. Attention $\xrightarrow[\text{similar.}]{\text{conceptually}}$ self Attention.

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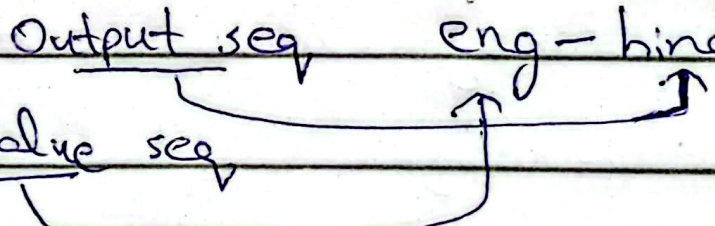
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Cross Attention:-

its a mechanism used in transformer architecture, particularly in task involving seq to seq data like translation or summarization. It allows a model to focus on different parts of input seq when generating output seq.

→ Cross attention needs both english & hindi embeddings unlike self which needs just embeddings.

→ Query → Output seq eng-hindi.
→ Key & value seq



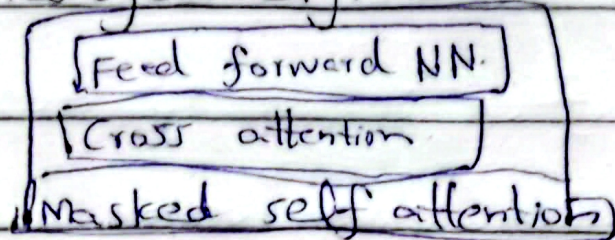
Uses:-

- image → Captioning. → text to image.
- text to speech.

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Decoder:- (Transformer):-



We divide it to 3 phases.

1) Input part 2) Processing 3) output

a) Input operation:-

i) Shifting (Just add start with sentence).

ii) Tokenizer (Break words into tokens or parts)

iii) Embedding (Convert words to nos.)

iv) Positional encoding (convert emb. vectors to unique vect)
for each pos by adding

b) Processing:-

After P.E our input goes to

mask multihead attention → Output

→ Add this output with the skip
connection output (Add and norm)

c)

2 Layer in F.F NN

1st layer = 2048 neuron $\rightarrow$ act relu

2nd " = 512 " $\rightarrow$ activation linear

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$\rightarrow$ Afterwards we do layer normalization.

$\rightarrow$ After it we do cross attention in which query vector come from this & key, value come from encoder

$\rightarrow$ (Add & norm) again this output & skip connection.

And normalize it.

$\rightarrow$ Now pass their output to feed forward NN

$\rightarrow$ Add N.N output with skip connection.

$\rightarrow$ Again apply layer normalization.

Now this output will go to 2nd decoder.

Output part:-

In output there is v no of neuron means according to unique words in sentences. - Apply linear function $\rightarrow$

$\rightarrow$ Apply softmax (every word have probability)

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→ Then find highest probability words.

- for that specific word → highest prob word will be output.

Transformer inference:-

In transformer encoder behaves as same both for training & inference.

- While decoder behaves differently as autoregressive.